Peptide toxicity prediction by fusing task-adaptive protein language model with pLDDT-aware multi-scale 3D structural modeling: GHXTox

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Abstract. Peptide toxicity is a major safety concern in peptide-based drug development, motivating accurate computational screening. Existing predictors often underuse three-dimensional (3D) structural information and provide limited cross-modal integration. We propose GHXTox, a hierarchical dual-level fusion framework that combines task-adaptive ProtT5–Bi-Mamba sequence modeling with pLDDT-aware multi-scale structural modeling. The structural expert integrates sequence semantics, residue geometry, and full-atom-derived chemical sites, and its probability is fused with that of the sequence expert using weights frozen from training out-of-fold predictions. On two frozen historical benchmarks, GHXTox achieved MCC values of 0.8705 and 0.6673 and AUPRC values of 0.9674 and 0.8046 on test set 1 and test set 2, respectively. The two experts provided complementary evidence, while residue- and structure-level analyses supported model interpretability. An OOF-frozen selective-prediction rule reduced accepted-sample error from 5.24% to 2.71% and from 5.33% to 3.16% at realized coverages of 88.54% and 86.94%, respectively. These retrospective results support confidence-aware sequence–structure fusion for peptide toxicity prediction and require confirmation on newly prefrozen external data.

Keywords: peptide toxicity prediction; multimodal deep learning; protein language model; bidirectional Mamba; pLDDT-aware multi-scale 3D modeling; interpretability

1 INTRODUCTION

Peptides are increasingly used as therapeutic candidates because of their high target specificity and functional diversity [1]. However, peptide toxicity remains a major safety concern during drug development, and the accurate and efficient identification of toxic peptides is a major bottleneck in peptide-based drug discovery [2]. Because conventional experimental toxicity assays are costly, time-consuming, and difficult to scale, reliable computational predictors are highly desirable [2].

Computational peptide toxicity prediction has evolved from traditional machine learning methods based on handcrafted physicochemical features toward deep learning approaches with stronger representation-learning capability. Current mainstream methods can be roughly grouped into two categories. The first category operates on sequence-level (one-dimensional, 1D) and/or atomic-level (two-dimensional, 2D) representations. For example, ToxMSRC combines continuous bag-of-words (CBOW) embeddings, multi-scale convolutional neural networks, and a bidirectional long short-term memory network with residual connections, achieving strong performance on the benchmark dataset [2]. ToxPLTC integrates pooled ProtT5 embeddings, borderline-SMOTE augmentation, and a text convolutional neural network [3]. ToxiPep fuses context-aware sequence representations with atomic-level graphs through a cross-attention mechanism [4]. HyPepTox-Fuse further combines embeddings from multiple protein language models with conventional descriptors [5]. Although these models achieve strong performance within their respective modalities, they largely neglect the three-dimensional (3D) conformations of peptides, which are closely related to toxicity; cross-dimensional information interaction and mutual validation remain limited, and their interpretability is mostly confined to the sequence level.

The second category attempts to exploit 3D structural information. PeptiTox uses ESM2 sequence embeddings together with ESMFold-predicted 3D structures and geometric graph neural networks [6]. StrucToxNet feeds ProtT5 sequence embeddings and ESMFold-predicted structures into an equivariant graph neural network (EGNN) and systematically analyzes structural quality [7]. These studies demonstrate the value of 3D conformations for toxicity prediction. However, these predictors still depend on predicted structures whose reliability can vary for short peptides, and both evaluate one predicted conformation per sequence rather than an explicit conformer ensemble. Existing designs use predicted-structure confidence primarily for filtering or structural-quality analysis rather than as a continuous message-level reliability signal [6–8]. Moreover, the fusion between the sequence and structure branches remains relatively shallow, and no systematic multi-dimensional fusion that combines task-adaptive sequence model with confidence-aware multi-scale structure modeling has been established.

To address these limitations, we propose GHXTox, an interpretable hierarchical dual-level fusion framework for peptide toxicity prediction. Task-adaptive 1D sequence expert combines a ProtT5 protein language model, adapted by low-rank adaptation (LoRA), with a bidirectional Mamba state-space encoder and three-seed ensembling [9–11]. A sequence-enhanced 3D structural expert integrates sequence semantics with spatial structure through pLDDT-aware cross-attention, ESMFold-derived geometry, full-atom chemical sites, and confidence gating across multiple structural scales; PepFlow and

AlphaFold2/ColabFold conformer ensembles were used to train and evaluate a separate residual head while the single-structure backbone remained frozen; this head is retained only for controlled ablations, and the final predictor uses the single-structure ESMFold anchor [12–15]. The two expert probabilities are fused at the probability level with weights fixed on training out-of-fold (OOF) predictions; the 1D probabilities are first calibrated on the same OOF table, and no benchmark labels are used during this selection. Gated sequence–structure cross-attention inside the structural expert further informs the interaction design [16]. Ablation experiments show that the contribution of structural evidence is dataset- and metric-dependent: fusion improves several fixed-threshold and probability-quality measures on test set 1 while maintaining broadly comparable performance on test set 2. Post hoc analyses characterize residue-level sensitivity, confidence gating, chemical-site perturbations, and expert disagreement without assigning causal biological meaning to these signals. The framework is evaluated on the frozen historical benchmarks described in Section 2.1.

In summary, the main contributions of this work are threefold: (1) we propose GHXTox, an interpretable hierarchical dual-level fusion framework that explores the complementary joint use of one-dimensional sequence and three-dimensional conformational information, integrating task-adaptive 1D protein language model with pLDDT-aware multi-scale 3D structure modeling and combining the two experts at the probability level (Sections 2.5 and 3.2); (2) the 3D branch designs a multi-scale structural information extraction strategy spanning from local residue functional groups to the whole peptide, and incorporates pLDDT confidence into structural feature processing through continuous down-weighting and dynamic gating; and (3) extensive experiments on the frozen historical benchmark test sets show that GHXTox improves several fixed-threshold and probability-quality measures on test set 1 while maintaining broadly comparable performance on test set 2, while ablation, robustness, and interpretability analyses characterize the conditions under which sequence and structural evidence are complementary.

2 MATERIALS AND METHODS

2.1 Datasets

We evaluate GHXTox on the benchmark datasets constructed by CAPTP [17] and subsequently adopted by ToxMSRC [2]. Following the data construction protocol of ToxMSRC [2], toxic peptide sequences were merged from three databases, namely CSMToxin [18], ToxinPred2 [19], and ATSE [20], and sequences longer than 50 residues were removed. Non-toxic sequences were retrieved from UniProt [21] using the query "NOT KW-0800 AND NOT KW-0020 AND reviewed:true", and sequences containing nonstandard amino acids or exceeding 50 residues were excluded. After combining the two groups and removing samples with conflicting labels, CD-HIT [22] was applied with a sequence-identity threshold of 0.9 to eliminate redundant sequences, yielding 2,138 toxic and 5,375 non-toxic peptides. Following the same partition as ToxMSRC [2], we used 85% of the data as the training set and the remaining 15% as benchmark test set 1, resulting in 1,818 toxic and 4,569 non-toxic training peptides and 320 toxic and 806 non-toxic test set 1 peptides. An additional benchmark, test set 2, containing 46 toxic and 536 non-toxic peptides, is used for frozen historical evaluation. Both test set 1 and

test set 2 had been observed during historical project development and are therefore treated as frozen historical benchmarks rather than newly blinded external validation sets; their labels were not used in the formal OOF calibration, fusion-weight selection, or threshold-freezing procedure. ToxinPred 3.0 [23] was consulted as an external method and data-source reference during project development, but its records were not added to the present training set or to the formal test set 1 and test set 2 result tables. Table 1 summarizes the statistics of the datasets used in this study.

2.2 Overview of the Model

GHXTox is a hierarchical dual-level fusion framework for peptide toxicity prediction: task-adaptive one-dimensional (1D) sequence expert and a sequence-enhanced three-dimensional (3D) structural expert generate independent probabilities, interact internally at the feature level, and are combined at the probability level.

The 1D expert processes the peptide sequence with a LoRA-adapted ProtT5 encoder [9, 10] followed by a bidirectional Mamba state-space model [11], averages three seed models, and applies OOF-fitted Platt calibration to produce p_{1D} (Section 2.3). The 3D expert integrates sequence semantics, residue geometry, full-atom chemical sites, and pLDDT confidence across residue, local-spatial, chemical-site, and peptide scales, combines sequence and spatial states through pLDDT-aware cross-attention, and emits p_{SE3D} (Section 2.4).

The two probabilities are combined at the probability level; the fusion rule, the decision threshold, and the final output are described in Section 2.5. The complete inference path is summarized in Figure 1.

2.3 Task-Adaptive 1D Sequence Modeling

The 1D sequence expert learns toxicity-related contextual semantics directly from peptide sequences. Instead of reducing a frozen protein-language-model output to a fixed mean vector, GHXTox preserves the residue-level representations of ProtT5, adapts the pretrained encoder to the toxicity task through low-rank adaptation (LoRA), models residue dependencies with a bidirectional Mamba encoder, and aggregates the residue states by masked mean pooling before classification. Three full-data models with different random seeds are combined by equal-weight probability averaging followed by Platt calibration fitted exclusively on training out-of-fold (OOF) predictions.

Each residue is first written in the space-separated format expected by the ProtT5 tokenizer. A binary residue-validity mask is constructed from the raw residue count, applied to invalid positions before each selective state-space scan, and reused in the final pooling; canonical inference follows the fixed batch-construction procedure used to generate the reported predictions. The tokenized sequence is processed by the ProtT5-XL-UniRef50 encoder [9] with LoRA, yielding residue-level representations that retain position-specific contextual semantics. To adapt this general-purpose encoder to the toxicity task, LoRA [10] is applied to all of its linear layers with rank $r = 8$, scaling $\alpha = 16$, and dropout 0.1: the pretrained weights stay frozen, and only the zero-initialized low-rank update ($\mathbf{W} = \mathbf{W}_0 + \frac{\alpha}{r}\mathbf{BA}$) and the downstream modules are trainable. The 1024-dimensional representations are then projected into the 192-dimensional Mamba hidden space through a layer-norm–GELU–dropout block.

Two bidirectional Mamba layers [11] are then stacked after the projection block to capture the sequential dependencies relevant to peptide toxicity. Each directional block follows the standard selective state-space formulation with input-dependent discretization and a 16-dimensional recurrent state; forward and backward scans are applied to the masked sequence, and their outputs are concatenated, fused by a linear layer, and added through a residual connection, so every residue is contextualized by both its N-terminal and C-terminal contexts. Invalid positions are masked before the selective scan and excluded from masked pooling; because the depthwise convolution precedes this scan mask in the implementation, the reported inference protocol retains the canonical batch construction. The resulting residue states are aggregated by masked mean pooling over the valid positions, and a layer-normalized single-output linear head yields the uncalibrated toxicity probability. Because non-toxic peptides far outnumber toxic peptides in the training data, a positive-class-weighted binary cross-entropy loss is used instead of SMOTE or Borderline-SMOTE, with the positive-class weight $w_+ = N_-/N_+$ recomputed from each fold's training partition; this emphasizes the minority class without generating synthetic samples or leaking held-out information.

To reduce the variance of a single training run, three 1D models are trained under identical settings with random seeds 42, 123, and 2025, and the ensemble score is the equal-weight arithmetic average of the three full-data models; no seed is selected on test performance and no seed-specific weight is introduced. Although averaging reduces variance, neural-network probabilities may remain miscalibrated, so a monotonic two-parameter Platt calibrator [24] is fitted to the training OOF scores (generated with the fixed group-aware five-fold partition, Section 2.5) and applied to the three-seed average at inference. Because the calibration is monotonic, it preserves the sample ranking. Labels from test set 1 and test set 2 are not used for member selection, calibration fitting, or threshold optimization. The calibrated probability $p_{1\mathrm{D}}$ is the sequence-expert output used in the top-level fusion (Section 2.5); no additional calibrator is fitted after fusion.

The complete 1D pipeline, from the peptide sequence to the calibrated probability, is summarized as follows:

$$\mathbf{H}^{\mathrm{PLM}} = \mathrm{ProtT5}_{\mathrm{LoRA}}(\mathcal{S}) \in \mathbb{R}^{L\times 1024}, \quad \mathbf{X}^{(0)} = \mathrm{Proj}\left[\mathbf{H}^{\mathrm{PLM}}\right] \in \mathbb{R}^{L\times 192}, \tag{1}$$

$$\mathbf{h}_{1D} = \frac{\sum_{i=1}^{L} m_i \ \mathrm{BiMamba}\left(\mathbf{X}^{(0)}\right)_i}{\sum_{i=1}^{L} m_i} \in \mathbb{R}^{192}, \tag{2}$$

$$z_{1D} = \mathbf{w}_c^{\top}\,\mathrm{LN}\left(\mathbf{h}_{1D}\right) + b_c, \qquad p_{1D}^{\mathrm{raw}} = \sigma\left(z_{1D}\right), \tag{3}$$

$$p_{1D} = \sigma\left(a\ \mathrm{logit}\left(\mathrm{clip}\left(\frac{1}{3}\sum_{k=1}^{3} p^{(k)},\ \varepsilon,\ 1-\varepsilon\right)\right) + b\right) \tag{4}$$

where Proj is the layer-norm–GELU–dropout projection, $\mathbf{m}$ is the residue-validity mask, $p^{(k)}$ is the probability of the k-th seed model, and a, b are the OOF-fitted monotonic Platt parameters (with logit clip $\varepsilon = 10^{-6}$).

2.4 pLDDT-Aware Multi-Scale 3D Structural Modeling

Peptide toxicity is influenced not only by amino acid composition and linear residue order but also by the three-dimensional (3D) arrangement of charged groups, hydropho-

bic regions, aromatic systems, and sulfur-containing moieties. Because peptides are generally short, conformationally flexible, and poorly covered by experimentally determined structures, treating a single predicted conformation as an error-free structural observation may amplify structure-prediction errors, whereas Cα-only or peptide-level summary representations fail to capture the local spatial-chemical environments that are relevant to toxicity [6–8].

To address these limitations, GHXTox employs a pLDDT-aware multi-scale structural expert that captures complementary single-structure information at four levels: residue semantics, spatial residue neighborhoods, full-atom chemical sites, and peptide-level global properties. Operationally, ESMFold provides a full-atom structural anchor and residue-level pLDDT values, while frozen ESM2 embeddings and explicit residue descriptors provide the internal sequence-semantic states. Cα geometry is encoded on a hybrid sequence/spatial-neighbor graph by a pLDDT-weighted equivariant network, and full-atom chemical pseudo-sites provide a zero-initialized residue-level correction. Contextual sequence states then query the confidence-gated spatial states through cross-attention; masked residue pooling and peptide-level physicochemical descriptors produce the structural logit. A separately trained, availability-masked conformer-ensemble residual can modify this logit only in controlled ablations and is disabled in the canonical benchmark and public-inference pathway.

Sections 2.4.1–2.4.6 specify this pipeline in order: single-structure prediction and conformer preparation, residue feature construction and the hybrid geometric graph, pLDDT-aware equivariant encoding, full-atom chemical-site message passing and the optional conformer residual, sequence–structure cross-attention, and global feature integration with classification. Throughout, pLDDT is treated as a continuous reliability signal rather than a hard filter. It enters the residue and edge features, weights geometric and chemical-site messages, and controls residue- and peptide-level structural gates. Low-confidence structural evidence is thereby down-weighted rather than removed, preserving structural coverage while limiting the influence of uncertain coordinates. pLDDT is interpreted here as a local confidence estimate produced by the structure-prediction model, not as an experimentally determined measure of structural accuracy, conformational flexibility, or biological activity.

2.4.1 Peptide Structure Prediction and Conformer Preparation

For a peptide sequence of length L, $S = (a_1, a_2, \ldots, a_L)$, ESMFold is first used to predict its three-dimensional structure [12]. The predicted structure is stored as a full-atom PDB file, from which residue-level Cα coordinates, backbone coordinates, full-atom-derived chemical-site coordinates, and pLDDT scores are extracted. The Cα coordinates are collected as $\mathbf{X} = [\mathbf{x}_1, \ldots, \mathbf{x}_L]^\top \in \mathbb{R}^{L\times 3}$, $\mathcal{A}$ denotes the full-atom-PDB-derived backbone and chemical-site geometry, and $\mathbf{q} = (q_1, \ldots, q_L) \in [0, 1]^L$ contains the residue-level pLDDT confidence scores rescaled to $[0, 1]$, where q_i denotes the residue-level pLDDT score of residue i. The Cα coordinates serve as the primary spatial variables of the residue-level equivariant graph, whereas the full-atom coordinates are used to construct local chemical sites. Before geometric encoding, the coordinates are centered at the peptide centroid to remove the effect of absolute translation.

A single predicted structure provides a common structural anchor but cannot repre-

sent potential conformational heterogeneity. The conformer-residual stage was therefore trained, with the single-structure spatial backbone, chemical-site branch, and base classifier frozen, on a multi-source conformer variant using a 2,795-peptide cohort with up to eight validated, aligned conformers per peptide. PepFlow is a generalized Boltzmann generator that performs direct all-atom conformational-ensemble sampling from the peptide energy landscape rather than predicting a single structure; it has been shown to efficiently recapitulate experimental ensembles of short linear motifs (SLiMs, typically 3–15 residues) and to generate all-atom conformations for peptides of different lengths [15]. Short-peptide conformations are therefore provided by PepFlow. In our pipeline, up to eight conformations are retained per input peptide, and PepFlow is applied to peptides of at most 15 residues (the actual training cohort spans 11–15 residues and comprises 761 records with eight conformers each). Longer peptides are provided by AlphaFold2/ColabFold: AlphaFold2 is a high-accuracy single-structure predictor [13], and its peptide-structure performance has been systematically evaluated for peptides of 16–60 residues [8], an interval that covers the 16–30-residue cohort; ColabFold, an MSA-accelerated AlphaFold2 implementation [14], is run in our pipeline with five models and multiple seeds to generate candidate conformations. These candidates are filtered by an average-pLDDT quality floor and then reduced to eight diverse conformers per peptide by farthest-point sampling on Kabsch-aligned Cα-RMSD [25], yielding 2,034 training records. These sources are represented explicitly in the residual head; nevertheless, generator identity remains partially coupled to peptide length and is treated as a limitation rather than as an independently identified causal factor.

For controlled ablation only, a test-side conformer cache was constructed with fixed source and sequence-matching policies without using test set 1 or test set 2 labels; these caches are not used by the final benchmark predictor or the public inference interface. To remove arbitrary rigid-body differences and make the statistics comparable across peptides, every admitted ensemble is normalized before being consumed by the residual branch. An ensemble is represented as $C = \{\mathbf{X}^{(1)}, \ldots, \mathbf{X}^{(M)}\}$, $1 \le M \le 8$, where $\mathbf{X}^{(m)} \in \mathbb{R}^{L\times 3}$ denotes the Cα coordinates of conformer m. First, each conformer must reproduce the input FASTA sequence residue by residue (both the Cα sequence and the chemical-site parse are verified); non-matching conformers are rejected, so all conformers share the same residue-index order without reordering or truncation. Second, taking the first conformer as the reference, every conformer is centered and aligned to the reference frame by Kabsch alignment, which obtains an optimal rotation by singular value decomposition with reflection correction, yielding the aligned coordinates and the per-residue fluctuation:

$$\begin{aligned} \tilde{\mathbf{X}}^{(m)} &= \left(\mathbf{X}^{(m)} - \mathbf{1}_L \bar{\mathbf{x}}^{(m)\top}\right)\mathbf{R}_m, \qquad \mathbf{R}_m = \mathbf{U}_m \mathbf{V}_m^{\top}, \\ \mathrm{RMSF}_i &= \left[\frac{1}{M}\sum_{m=1}^{M}\left\|\tilde{\mathbf{x}}_i^{(m)} - \bar{\mathbf{x}}_i\right\|_2^2\right]^{1/2}, \end{aligned} \tag{5}$$

where $\tilde{\mathbf{X}}^{(m)}$ denotes the aligned Cα coordinates of conformer m, $\mathbf{1}_L$ is the all-ones vector, $\bar{\mathbf{x}}^{(m)}$ is the centroid of conformer m, $\mathbf{R}_m$ is the optimal rotation obtained from the singular value decomposition of the centered cross-covariance between conformer

m and the reference conformer (with reflection correction), $\bar{\mathbf{x}}_i$ is the ensemble-mean coordinate of residue i, and RMSF_i is the corresponding per-residue root-mean-square fluctuation.

After alignment, the ensemble-mean coordinates, per-residue root-mean-square fluctuation (RMSF), and per-conformer RMSD are computed. Rotation- and translation-invariant quantities, including per-conformer radius of gyration, per-residue-pair distance mean and standard deviation, and 6/8/10 Å contact occupancies (the fraction of conformers in which a residue pair is closer than the cutoff) – are computed directly from the raw coordinates. The resulting statistics are rescaled by fixed factors (e.g., RMSF/10 and distances/20) into bounded residue-level (8-dimensional) and peptide-level (12-dimensional) features, and the conformer count is normalized as $M/8$. Regarding confidence, AlphaFold2/ColabFold conformers carry the real pLDDT read from the PDB B-factor column, whereas PepFlow does not output pLDDT and is assigned a fixed substitute value of 0.7; this substitute is explicitly not treated as pLDDT in the single-structure quality gating, which remains governed by the ESMFold backbone, and PepFlow contributes only cross-conformer fluctuation and occupancy information. The normalized ensemble statistics are consumed by a small logit residual branch controlled by a binary availability indicator c_{conf} (mechanism detailed in Section 2.4.4); when $c_{\mathrm{conf}} = 0$, the correction vanishes exactly and the model reduces to the single-structure pathway. Missing conformer statistics are treated as unavailable data rather than as evidence of structural rigidity; the model draws no flexibility inference from the absence of conformers and simply falls back to the single-structure prediction.

The generated conformers are computational samples rather than experimentally determined trajectories or rigorous thermodynamic ensembles. Consequently, RMSF, contact occupancy, and distance variance are interpreted as sampling-derived descriptors of conformational variation rather than direct measurements of peptide dynamics in solution.

2.4.2 Residue Feature Construction and Hybrid Geometric Graph

Given the predicted single structure and its confidence scores (Section 2.4.1), the structural expert constructs two complementary residue-level representations: an internal sequence-semantic representation and a spatial-geometric representation. The sequence-semantic subbranch uses frozen ESM2 residue embeddings [12] and a lightweight Transformer; unlike the independent top-level 1D prediction expert of Section 2.3, it serves the residue-level sequence–structure alignment inside the structural expert.

For residue i, the input to the internal sequence-semantic subbranch is

$$\mathbf{u}_i^{\mathrm{seq}} = \left[\mathbf{e}_i^{\mathrm{AA}} \parallel \mathbf{e}_i^{\mathrm{group}} \parallel \mathbf{f}_i^{\mathrm{res}} \parallel P_{\mathrm{ESM}}\left(\mathbf{e}_i^{\mathrm{ESM2}}\right)\right], \tag{6}$$

where $\mathbf{e}_i^{\mathrm{AA}} \in \mathbb{R}^{32}$ is the amino acid category embedding, $\mathbf{e}_i^{\mathrm{group}} \in \mathbb{R}^{12}$ is the functional-group category embedding, $\mathbf{f}_i^{\mathrm{res}} \in \mathbb{R}^{42}$ denotes the explicit residue-level physicochemical descriptors, and $\mathbf{e}_i^{\mathrm{ESM2}} \in \mathbb{R}^{1280}$ is the ESM2-650M residue representation; the projection P_{ESM} maps the ESM2 representation to 128 dimensions. Stacking the residue-level feature vectors yields the sequence-feature matrix $U_{\mathrm{seq}} \in \mathbb{R}^{L\times 214}$ ($214 = 32 + 12 + 42 + 128$). After a linear projection and positional encoding, a two-layer Transformer

encoder (hidden size 128, four attention heads, feed-forward dimension 512) produces the contextualized sequence states $\mathbf{H}^{\mathrm{seq}} \in \mathbb{R}^{L\times 128}$, which encode the contextual sequence semantics of each residue and serve as the query representation in the sequence–structure cross-attention module [26].

In the spatial subbranch, the initial representation of residue i is

$$\mathbf{h}_i^{(0)} = P_{\mathrm{sp}}\left[\mathbf{f}_i^{\mathrm{res}} \parallel \mathbf{g}_i^{\mathrm{geo}} \parallel q_i \parallel P_{\mathrm{ESM}}\left(\mathbf{e}_i^{\mathrm{ESM2}}\right)\right], \tag{7}$$

where $\mathbf{g}_i^{\mathrm{geo}} \in \mathbb{R}^{16}$ is a lightweight geometric descriptor derived from the Cα coordinates and pLDDT scores, including (i) residue-level pLDDT and local confidence statistics; (ii) radial position, contact fractions, and soft spatial density; (iii) nearest-neighbor, nonlocal-neighbor, and local sequential distances; and (iv) sine/cosine encodings of the Cα-trace angles and torsions. The residue-level confidence q_i additionally enters the initial representation as an explicit scalar. These deterministic descriptors provide stable local geometric priors without imposing an overly complex handcrafted representation for short and uncertain predicted peptide structures.

GHXTox then constructs a hybrid residue graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}_{\mathrm{seq}} \cup \mathcal{E}_{\mathrm{knn}})$ in which each residue corresponds to one node: sequential edges $\mathcal{E}_{\mathrm{seq}}$ connect adjacent residues and preserve peptide-chain continuity, whereas $\mathcal{E}_{\mathrm{knn}}$ connects each residue to its 10 nearest spatial neighbors and is symmetrized. The hybrid graph therefore captures both covalent sequence order and nonlocal spatial contacts while avoiding large variations in graph density that may arise from a single fixed-distance cutoff; equivalently, $A_{ij} \in \{0, 1\}$ indicates whether residues i and j are connected by a sequential or k-nearest-neighbor edge.

For residues i and j, the Cα distance $d_{ij} = \|\mathbf{x}_i - \mathbf{x}_j\|_2$ is encoded by 16 radial basis functions with centers μ_k uniformly spaced over 0–24 Å, $\phi_k(d_{ij}) = \exp[-\gamma(d_{ij} - \mu_k)^2]$ with $\gamma = 1/(\mu_{k+1} - \mu_k)^2$. The base edge representation additionally includes the pairwise structural confidence and the normalized sequence separation:

$$c_{ij} = \sqrt{q_i q_j}, \qquad s_{ij} = \frac{\log(1 + |i - j|)}{\log(L + 1)}, \tag{8}$$

so that s_{ij} is small for sequentially adjacent residues and approaches one for distant residues. The resulting edge feature is therefore $\mathbf{e}_{ij} = \left[\boldsymbol{\phi}(d_{ij}) \parallel c_{ij} \parallel s_{ij}\right]$, which lets the model distinguish sequentially adjacent contacts, long-range spatial contacts, and contacts supported by different levels of structural confidence.

2.4.3 pLDDT-Aware Equivariant Graph Encoding

To learn three-dimensional representations without relying on a particular global coordinate frame, the spatial branch employs three E(3)-equivariant graph neural network layers [27]. The network updates residue features using rotation- and translation-invariant quantities while updating coordinates equivariantly. Consequently, a global rigid transformation of the peptide structure does not change the predicted toxicity, whereas intermediate coordinate representations transform consistently with the input.

At layer ℓ, the message from residue j to residue i is computed as $\mathbf{m}_{ij}^{(\ell)} = \phi_e^{(\ell)}[\mathbf{h}_i^{(\ell)} \parallel \mathbf{h}_j^{(\ell)} \parallel \mathbf{e}_{ij}^{(\ell)}]$, where $\phi_e^{(\ell)}$ is a message multilayer perceptron. To prevent uncertain

or distant contacts from contributing as strongly as local high-confidence contacts, a continuous edge weight is defined as

$$w_{ij}^{(\ell)} = A_{ij}\,\exp\!\left(-\frac{d_{ij}^{(\ell)}}{12}\right)\left[q_{\min} + (1 - q_{\min})\,c_{ij}^{\rho}\right], \qquad q_{\min} = 0.1, \quad \rho = 1.5, \tag{9}$$

The term $\exp(-d_{ij}^{(\ell)}/12)$ imposes a smooth distance decay with a characteristic decay length of 12 Å, and both the distance and the edge weight are recomputed at each layer from the current coordinates. Here, $A_{ij} \in \{0, 1\}$ is the adjacency indicator of the hybrid graph defined in Section 2.4.2, and $c_{ij} = \sqrt{q_i q_j}$ is the pairwise structural confidence derived from the residue-level pLDDT scores q_i and q_j defined in Section 2.4.1. The minimum weight $q_{\min}$ prevents low-confidence regions from being completely disconnected, whereas the exponent ρ increases the relative contribution of high-confidence contacts. The distance-decay term further reduces the influence of more distant spatial neighbors.

These pLDDT-aware weighted edge messages are then aggregated at each node with a normalization floor. For node i, the aggregated message is

$$\mathbf{m}_i^{(\ell)} = \frac{\sum_j w_{ij}^{(\ell)} \mathbf{m}_{ij}^{(\ell)}}{\max\left(\sum_j w_{ij}^{(\ell)}, 1\right)}, \tag{10}$$

and the node features are updated as

$$\mathbf{h}_i^{(\ell+1)} = \mathrm{LN}\left\{\mathbf{h}_i^{(\ell)} + \phi_h^{(\ell)}\left[\mathbf{h}_i^{(\ell)} \parallel \mathbf{m}_i^{(\ell)}\right]\right\}. \tag{11}$$

The coordinates are updated equivariantly as

$$\mathbf{x}_i^{(\ell+1)} = \mathbf{x}_i^{(\ell)} + 0.05\,\frac{\sum_j \alpha_{ij}^{(\ell)}\left(\mathbf{x}_i^{(\ell)} - \mathbf{x}_j^{(\ell)}\right)}{\max\left(\sum_j w_{ij}^{(\ell)}, 1\right)}, \tag{12}$$

where

$$\alpha_{ij}^{(\ell)} = \tanh\left[\phi_x^{(\ell)}\left(\mathbf{m}_{ij}^{(\ell)}\right)\right] w_{ij}^{(\ell)} \tag{13}$$

is a bounded coordinate-update coefficient generated from the edge message and pairwise distances. The coordinates are recentered after every layer to reduce numerical drift. Because the update depends only on scalar messages, distances, and relative coordinate vectors, the operation remains equivariant to rigid rotations and translations.

Each layer update performs one round of message passing, i.e., one hop of spatial-context propagation over the hybrid graph, so a single layer can aggregate only the direct neighborhood of each node. Because toxicity-related spatial patterns often span multiple residues (for example, spatial clusters formed by several charged, aromatic, or hydrophobic residues), the model must aggregate multi-hop neighborhood information along the graph; stacking layers therefore enlarges the receptive field hop by hop: the first layer primarily encodes direct geometric neighbors, whereas subsequent layers integrate higher-order spatial environments through repeated message passing. GHXTox

uses three layers, a depth that balances context integration against model complexity and overfitting risk on short and uncertain predicted peptide structures. Unlike approaches that simply append pLDDT as an ordinary node feature, GHXTox uses pLDDT to directly modulate the strength of geometric messages, allowing structural uncertainty to influence representation learning at the propagation stage. Low pLDDT does not imply that a residue is biologically unimportant; it may reflect prediction error or genuine conformational flexibility, so the model reduces the reliability of uncertain coordinate evidence while preserving the residue's sequence-semantic and physicochemical features.

2.4.4 Full-Atom Chemical-Site Message Passing

A Cα-based residue graph captures the global fold and spatial neighborhoods but cannot resolve chemically distinct moieties within the same side chain. For example, tyrosine contains both an aromatic ring and a hydroxyl group, whereas tryptophan contains an aromatic system and an indole nitrogen. Compressing each side chain into a single centroid would mix these chemically different moieties. GHXTox therefore extracts chemistry-aware pseudo-sites from the predicted full-atom PDB structure and processes them with a dedicated local message-passing branch [28].

To retain this chemical detail, up to two chemical sites are kept for each residue. Site identity is represented by an eight-dimensional multi-hot vector $\mathbf{t}_{ir} \in \{0,1\}^8$ allowing overlapping types: positive, negative, donor, acceptor, aromatic, hydrophobic, sulfur, and conformational, where r indexes the chemical site within residue i. Multiple types may be assigned to the same site: for example, the cysteine sulfur site is labeled as donor, hydrophobic, and sulfur-containing, whereas the proline ring is labeled as hydrophobic and conformationally constrained; the aromatic ring and hydroxyl group of tyrosine are represented as separate sites, as are the aromatic system and the indole NE1 nitrogen (donor) of tryptophan.

Each site is further localized and oriented in the predicted structure: its coordinate is the mean position of the heavy atoms defining the corresponding chemical moiety, and its orientation is represented by either a bond direction or an unsigned plane normal. Because the sign of a plane normal is arbitrary, downstream directional features use absolute cosines, preserving rotation invariance and eliminating orientation-sign ambiguity. Combining these attributes, the initial representation of site r in residue i is $\mathbf{z}_{ir}^{(0)} = \phi_{\text{site}}[\mathbf{t}_{ir} \parallel q_i \parallel o_{ir}^{\text{valid}}]$, where q_i is the pLDDT score of the parent residue and o_{ir}^{valid} indicates whether a reliable orientation could be constructed; the input is projected into a 64-dimensional hidden space.

Because chemical interactions depend on spatial proximity, edges are established between valid sites separated by no more than 16 Å; same-residue site edges are kept, preserving spatial relationships between chemically different moieties in complex side chains. Each edge carries four complementary feature families. First, the raw distance in Å is encoded using 16 radial basis functions, $\boldsymbol{\phi}_{\text{raw}}(d_{uv}) \in \mathbb{R}^{16}$. Second, a peptide-size-normalized distance is constructed. The radius of gyration of the Cα coordinates

is

$$R_g = \sqrt{\frac{1}{L}\sum_{i=1}^{L} \|\mathbf{x}_i - \bar{\mathbf{x}}\|_2^2}, \qquad \bar{\mathbf{x}} = \frac{1}{L}\sum_{i=1}^{L} \mathbf{x}_i, \qquad \bar{d}_{uv} = \frac{d_{uv}}{R_g}, \tag{14}$$

where $\bar{\mathbf{x}}$ denotes the centroid of the valid Cα coordinates, and the normalized distance is encoded by eight radial basis functions over the interval 0–4, $\boldsymbol{\phi}_{\mathrm{norm}}(\bar{d}_{uv}) \in \mathbb{R}^8$. The raw representation describes the physical separation in Å, whereas the normalized representation describes the separation relative to the overall peptide size, allowing the model to distinguish, for example, an 8 Å contact in a compact short peptide from an 8 Å contact in a larger peptide. Third, four rotation-invariant orientation features are used, $\mathbf{o}_{uv} = [|\mathbf{o}_u \cdot \mathbf{r}_{uv}|, |\mathbf{o}_v \cdot \mathbf{r}_{vu}|, |\mathbf{o}_u \cdot \mathbf{o}_v|, o_{uv}^{\mathrm{valid}}]$, which describe the alignment of the source orientation with the intersite direction, the alignment of the target orientation with the reverse direction, the mutual orientation of the two sites, and orientation availability. Fourth, six explicit compatibility channels are used: opposite-charge, donor–acceptor, aromatic–aromatic, positive–aromatic (cation–π), sulfur–sulfur, and hydrophobic–hydrophobic compatibility. These channels encode candidate chemical compatibility patterns available to the model; they do not imply that a true salt bridge, hydrogen bond, cation–π interaction, disulfide bond, or hydrophobic core is present in the predicted structure.

Combining these families yields the full chemical-site edge representation

$$\mathbf{e}_{uv}^{\mathrm{chem}} = \left[\boldsymbol{\phi}_{\mathrm{raw}}(d_{uv}) \parallel \boldsymbol{\phi}_{\mathrm{norm}}(\bar{d}_{uv}) \parallel \mathbf{o}_{uv} \parallel \mathbf{c}_{uv}^{\mathrm{chem}} \parallel \sqrt{q_u q_v}\right], \tag{15}$$

in which pLDDT enters the chemical-site features as a confidence channel. At layer l, each site message is computed from the two site states and the edge representation, weighted by the pairwise confidence, and aggregated over the neighborhood. After two message-passing layers, valid sites belonging to the same residue are mean-pooled, and the residue-level chemical residual and the chemical-enhanced spatial state are

$$\begin{aligned}
\mathbf{m}_{uv}^{(l)} &= \phi_{\mathrm{chem}}^{(l)}\left[\mathbf{z}_u^{(l)} \parallel \mathbf{z}_v^{(l)} \parallel \mathbf{e}_{uv}^{\mathrm{chem}}\right], \\
\bar{\mathbf{m}}_v^{(l)} &= \frac{\sum_{u\in\mathcal{N}(v)} \sqrt{q_u q_v}\, \mathbf{m}_{uv}^{(l)}}{\max\left(\sum_{u\in\mathcal{N}(v)} \sqrt{q_u q_v},\, 1\right)}, \\
\Delta\mathbf{h}_i^{\mathrm{chem}} &= \mathbf{W}_{\mathrm{chem}} \operatorname*{Mean}_{r\in\mathcal{S}_i}\left(\mathbf{z}_{ir}^{(2)}\right), \\
\mathbf{h}_i^{\mathrm{sp}} &= \mathbf{h}_i^{\mathrm{EGNN}} + \Delta\mathbf{h}_i^{\mathrm{chem}},
\end{aligned} \tag{16}$$

where the aggregation weights the incoming messages by the pairwise confidence $\sqrt{q_u q_v}$, continuing the pLDDT-aware weighting of information extraction. The projection $\mathbf{W}_{\mathrm{chem}}$ and its bias $\mathbf{b}_{\mathrm{chem}}$ are zero-initialized, so that $\Delta\mathbf{h}_i^{\mathrm{chem}} = 0$ at the beginning of training. This conservative residual design allows the new chemical-site branch to progressively learn useful corrections from an intact geometric backbone rather than perturbing the existing spatial representation with a randomly initialized contribution.

Beyond these residue-level corrections, the structural expert additionally allows multi-conformer statistics to refine the prediction through a logit-level residual [29]. To avoid the high cost of running the complete geometric network independently on

every conformer, GHXTox compresses the ensemble descriptors normalized in Section 2.4.1 into a lightweight statistical residual: residue-level ensemble features are encoded and pooled with both masked-mean and masked-max operations, the peptide-level descriptor is processed by a separate encoder, and the generator identity (PepFlow, AlphaFold2/ColabFold, or another admitted generator) is encoded by a three-dimensional one-hot vector whose learned source embedding is concatenated with the base structural embedding and the two ensemble embeddings to produce a raw logit residual through a small network with a zero-initialized final layer. The effective residual is the product of the availability indicator and a learned weight (initialized to 0.25 and regularized by an L1 penalty); when the availability indicator is zero, the residual is exactly zero and the model falls back to the single-structure prediction. The inference protocol is described in Section 2.5, and the ablation analysis in Section 3.3.

2.4.5 pLDDT-Aware Sequence–Structure Cross-Attention

After the EGNN and the chemical-site residual are applied, the model obtains the chemical-enhanced spatial representation $\mathbf{H}^{\mathrm{sp}} = [\mathbf{h}_1^{\mathrm{sp}}, \ldots, \mathbf{h}_L^{\mathrm{sp}}]^\top \in \mathbb{R}^{L\times 128}$, where $\mathbf{h}_i^{\mathrm{sp}}$ is defined in Eq. (16). Spatial information alone does not determine whether a structural pattern is relevant to the current sequence context: similar hydrophobic contacts may have different implications depending on residue identities and sequence positions. GHXTox therefore uses pLDDT-aware cross-attention within the structural expert, allowing contextualized sequence states to selectively retrieve spatial evidence.

A node-level gate is first constructed for each residue from the 42-dimensional residue descriptors and the pLDDT score:

$$
\begin{aligned}
g_i^{\mathrm{base}} &= \phi_{\mathrm{node}}\left[\mathbf{f}_i^{\mathrm{res}} \parallel q_i\right], \\
\mathbf{c}_i^{\mathrm{node}} &= \left[\frac{q_i-\mu_q}{T_q},\ \frac{(q_i-\mu_q)^2}{T_q^2},\ \mathbb{I}(q_i<0.55)\right], \\
r_i^{\mathrm{node}} &= 0.5 + \sigma\left(\mathbf{w}_n^\top \mathbf{c}_i^{\mathrm{node}} + b_n\right), \\
g_i^{\mathrm{node}} &= \mathrm{clamp}\left(g_i^{\mathrm{base}}\, r_i^{\mathrm{node}},\ 0, 1\right),
\end{aligned}
\tag{17}
$$

where $\mu_q = 0.70$, $T_q = 0.12$, and the residual confidence vector $\mathbf{c}_i^{\mathrm{node}}$ allows the gate to adapt to the local confidence distribution. The spatial representation is then continuously down-weighted as $\widehat{\mathbf{H}}^{\mathrm{sp}} = \mathbf{H}^{\mathrm{sp}} \odot \mathbf{g}^{\mathrm{node}}$, with the node-level gate vector $\mathbf{g}^{\mathrm{node}} \in [0,1]^L$ broadcast across the feature dimension.

In the cross-attention operation, the contextualized sequence representation serves as the query, and the gated spatial representation serves as the key and value [26]:

$$
\mathbf{H}^{\mathrm{cross}} = \mathrm{MHA}\left(Q = \mathbf{H}^{\mathrm{seq}}, K = \widehat{\mathbf{H}}^{\mathrm{sp}}, V = \widehat{\mathbf{H}}^{\mathrm{sp}}\right). \tag{18}
$$

This direction has a direct interpretation: each residue position retrieves spatial evidence according to its contextual sequence representation. The operation is not a direct feature interaction between the LoRA-ProtT5–Mamba expert and the structural expert; it occurs internally between the ESM2-enhanced sequence states and the geometric states of the structural expert.

In addition to residue-level gating, a peptide-level gate is generated from the overall structure-confidence profile. The model first summarizes the profile by the mean pLDDT $\bar{q}$, the minimum pLDDT $q_{\min}$, and the low-confidence fraction $r_{\text{low}} = \frac{1}{L}\sum_{i=1}^{L}\mathbb{I}(q_i < 0.55)$, and constructs the peptide-level confidence vector $\mathbf{c}^{\text{pep}} = \left[(\bar{q}-\mu_q)/T_q,\ (q_{\min}-\mu_q)/T_q,\ r_{\text{low}}\right]$. The base peptide-level gate and its residual-adjusted form are

$$\begin{aligned} g_{\text{pep}}^{\text{base}} &= \phi_{\text{global}}\left[\bar{q},\ q_{\min},\ r_{\text{low}}\right], \\ r_{\text{pep}} &= 0.5 + \sigma\left(\mathbf{w}_{\text{pep}}^{\top}\mathbf{c}^{\text{pep}} + b_{\text{pep}}\right), \\ g_{\text{pep}} &= \text{clamp}\left(g_{\text{pep}}^{\text{base}}\, r_{\text{pep}},\ 0, 1\right), \end{aligned} \tag{19}$$

where the peptide-level residual scale is analogous to, but separately parameterized from, the node-level residual scale, and $g_{\text{pep}} \in [0, 1]$ denotes the scalar peptide-level structural gate.

The final residue-level fusion representation combines the sequence state with the gated structural context, $\mathbf{H}^{\text{fused}} = \text{LN}\left[\mathbf{H}^{\text{seq}} \parallel g_{\text{pep}}\,\mathbf{H}^{\text{cross}}\right]$. The two gating levels answer complementary questions: how reliable the structural evidence is at an individual residue position, and how strongly the structural branch should be trusted for the peptide as a whole. The sequence representation is always kept, whereas the structural context is dynamically scaled according to prediction confidence. Consequently, when the predicted structure has low overall confidence, the structural expert can still rely on its internal sequence-semantic pathway without allowing uncertain coordinates to dominate the classification.

2.4.6 Global Feature Integration and Structural Classification

After pLDDT-aware cross-attention, each valid residue is represented by a 256-dimensional fused vector obtained by concatenating the 128-dimensional internal sequence state with the 128-dimensional structural context. Masked mean pooling is applied over the valid residue positions, and the pooled vector is concatenated with five global physicochemical descriptors before a three-layer MLP produces the base structural logit; the availability-masked conformer correction described in Section 2.4.4 is then added, and a sigmoid yields the final structural-expert probability

$$p_{\text{SE3D}} = \sigma\left(\phi_{\text{cls}}\left[\left(\frac{\sum_{i=1}^{L} m_i\,\mathbf{H}_i^{\text{fused}}}{\sum_{i=1}^{L} m_i}\right) \parallel \mathbf{f}_{\text{pep}}^{\text{phys}}\right] + c_{\text{conf}}\,\lambda_{\text{conf}}\,\Delta z_{\text{conf}}^{\text{raw}}\right), \tag{20}$$

where the mask excludes padding positions from the pooled representation; $\mathbf{f}_{\text{pep}}^{\text{phys}} \in \mathbb{R}^5$ contains the normalized peptide length, the length-normalized net charge, the aromatic-residue fraction, the cysteine fraction, and the scaled mean hydropathy; ϕ_{cls} is a three-layer MLP with dimensions $261 \to 128 \to 64 \to 1$, each hidden layer applying GELU activation and dropout; and the conformer-correction term is nonzero only when valid ensemble information is available (Section 2.4.4), so under the benchmark and public inference protocol $c_{\text{conf}} = 0$ and p_{SE3D} reduces to $\sigma(z_{\text{base}})$, i.e., the single-structure base logits. Unlike maximum pooling, masked mean pooling integrates structural evidence distributed across residue positions, which is appropriate when toxicity-related spatial

patterns are not confined to a single local region. The resulting probability is passed directly to the dual-expert fusion in Section 2.5; the standalone structural threshold is used only for branch-specific analyses.

2.5 Fusion and Final Output

The 1D expert emits the OOF-calibrated probability p_{1D} (Section 2.3), and the structural expert emits p_{SE3D} (Section 2.4). The two probabilities are fused linearly in probability space with weights frozen on training out-of-fold (OOF) predictions:

$$p_{GHXTox} = 0.72\, p_{1D} + 0.28\, p_{SE3D}, \tag{21}$$

The fused probability p_{GHXTox} is the model's continuous toxicity-probability output; it is binarized directly at the OOF-frozen threshold, predicting toxic ($\hat{y} = 1$) if $p_{GHXTox} \geq 0.47022256531284556$ and non-toxic ($\hat{y} = 0$) otherwise. The default output is therefore $(p_{GHXTox}, \hat{y})$. Experiments showed that directly emitting the fused probability yields the best overall performance, so no additional network or calibration is applied after fusion.

The two experts were trained independently because they differ in input representation, parameter scale, and optimization behavior (the 1D and 3D loss, calibration, and training details are described in Sections 2.3 and 2.4). Group-aware training OOF predictions were used for calibration, fusion-weight selection, and threshold determination, and labels from test set 1 and test set 2 were excluded from all of these procedures. The training sequences were clustered by CD-HIT at a sequence-identity threshold of 0.80, with all peptides in the same homology group confined to the same fold; each fold was predicted by a model trained only on its complement, giving every training peptide a single OOF probability independent of its own optimization [22]. Both experts were optimized with AdamW [30]. The fusion weight and decision threshold were selected jointly on the complete group-aware training OOF table by maximizing MCC; in exact MCC ties, the larger threshold was retained. This procedure yielded $w_{1D} = 0.72$, $w_{SE3D} = 0.28$, and $\tau = 0.47022256531284556$; these values were then frozen and applied unchanged to test set 1 and test set 2. In the final benchmark and public inference protocol, the conformer availability mask is fixed to zero and the structural probability is read from the single-structure base logits.

2.6 Evaluation Criteria

Given the class-imbalanced benchmarks (test set 2 has 46 toxic and 536 non-toxic peptides), MCC and BACC served as the primary threshold-dependent metrics, with ACC, sensitivity (SN), specificity (SP), precision, and F1 reported for completeness. AUROC and AUPRC quantified ranking performance; AUPRC complements AUROC under class imbalance by emphasizing precision–recall behavior for the positive class [31]. Brier/ECE-10 assessed probability quality. Threshold-dependent metrics used the fixed OOF-derived threshold (Section 2.5); ranking and calibration metrics used the continuous probabilities. Binary predictions were obtained as

$$\hat{y}_i = \mathbb{I}[p_i \geq \tau]\,. \tag{22}$$

The threshold-dependent performance measures were summarized by

$$\mathrm{ACC} = \frac{\mathrm{TP} + \mathrm{TN}}{N}, \tag{23}$$

$$\mathrm{SN} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}, \tag{24}$$

$$\mathrm{SP} = \frac{\mathrm{TN}}{\mathrm{TN} + \mathrm{FP}}, \tag{25}$$

$$F_1 = \frac{2\mathrm{TP}}{2\mathrm{TP} + \mathrm{FP} + \mathrm{FN}}, \tag{26}$$

$$\mathrm{BACC} = \frac{\mathrm{SN} + \mathrm{SP}}{2}. \tag{27}$$

MCC [32], Brier [33], and ten-bin ECE-10 [34] (final bin includes $p_i = 1$) are

$$\begin{aligned} \mathrm{MCC} &= \frac{\mathrm{TP}\,\mathrm{TN} - \mathrm{FP}\,\mathrm{FN}}{\sqrt{(\mathrm{TP} + \mathrm{FP})(\mathrm{TP} + \mathrm{FN})(\mathrm{TN} + \mathrm{FP})(\mathrm{TN} + \mathrm{FN})}}, \\ \mathrm{Brier} &= \frac{1}{N}\sum_{i=1}^{N}(p_i - y_i)^2, \\ \mathrm{ECE}_{10} &= \sum_{m=1}^{10}\frac{|\mathcal{B}_m|}{N}\,\left|\mathrm{conf}(\mathcal{B}_m) - \mathrm{freq}(\mathcal{B}_m)\right|, \end{aligned} \tag{28}$$

where TP/TN/FP/FN are the true/false positive/negative counts, y_i the labels, and $\mathcal{B}_m$ the m-th equal-width bin over $[0, 1]$ (mean confidence $\mathrm{conf}(\mathcal{B}_m)$ vs. observed toxic frequency $\mathrm{freq}(\mathcal{B}_m)$); ECE bins stayed fixed across models and datasets.

Uncertainty intervals and paired model comparisons use bootstrap resampling [35], retaining the same peptides across paired methods; the applicable stratification and replicate count are stated with each analysis. In the expert-complementarity analysis, reciprocal rescue rates use 95% Wilson score intervals [36]. Continuous attribute effects are summarized by Cliff's delta [37]; their intervals use 5,000 label-stratified bootstrap replicates, while p-values use 10,000 true-label-stratified permutations [38]. The ten prespecified split-by-attribute tests form one family, for which the Benjamini–Hochberg procedure controls the false discovery rate [39].

3 RESULTS AND DISCUSSION

3.1 Overall Predictive Performance on the Benchmark Test Sets

Using the OOF-frozen weight and threshold without dataset-specific retuning, GHXTox correctly classified 1,067 of 1,126 test set 1 peptides (TP/TN/FP/FN = 287/780/26/33) and 551 of 582 strongly imbalanced test set 2 peptides (35/516/20/11); the complete metric set is listed in Table 2. The low test set 2 BACC (0.8618) and F1 (0.6931), despite high accuracy, reflect the difficulty of recovering the 46 toxic peptides and demonstrate why class-balanced and precision–recall metrics are essential for this benchmark.

Table 2 also reports previously published models on the same benchmark partitions (ToxMSRC and ToxPLTC; values transcribed from the original publications, and NR denotes an unreported metric) [2, 3]. GHXTox obtains the highest test set 1 ACC,

BACC, SP, MCC, AUROC, and AUPRC among the compared rows, whereas ToxPLTC has higher test set 1 SN; on test set 2, GHXTox obtains the highest ACC, SP, MCC, and reported AUPRC, but its BACC and SN remain below both reference models. This reflects a stronger overall balance of specificity, correlation, and positive-class ranking rather than comprehensive superiority on every metric. Because test set 1 and test set 2 were observed historically, these differences remain retrospective benchmark evidence and require confirmation on a new prefrozen external set.

The final fused probabilities have Brier scores of 0.0410 and 0.0436 and ECE-10 values of 0.0196 and 0.0495 on test set 1 and test set 2, respectively. The test set 2 AUPRC (0.8046) directly reflects the precision–recall trade-off for the rare toxic class, which is not conveyed by AUROC alone [31], and the 11 false negatives on test set 2 delineate the limits of the operating point: high specificity should not be interpreted as complete minority-class coverage. Discrimination and calibration curves are shown in Figure 2.

3.2 Complementarity of the 1D and 3D Experts

The two experts are fused only after independent probability generation. The 1D branch provides the stronger standalone ranking signal, whereas the 3D branch contributes confidence-weighted geometry and local chemical context. This separation makes the gain attributable to complementary probability evidence rather than duplicated use of an earlier fused prediction (Table 3).

The experts agreed on most benchmark peptides, while discordant cases revealed class-asymmetric correction patterns (Figure 3).

Relative to the calibrated 1D probability stream, the final fusion modestly improves AUROC/AUPRC and lowers the Brier score on both benchmarks (Table 3), adding complementary structural ranking information while maintaining a lower probability error. The structural probability in these rows is the single-structure base output of the multi-source-trained structural checkpoint; no benchmark conformer cache is activated.

The 0.72 sequence weight is an empirical result of the training-OOF search, not a manually imposed assertion that sequence information is intrinsically more important. For the current canonical evaluation, both expert weights and the full-precision decision threshold of 0.47022256531284556 were frozen from training OOF predictions and applied without benchmark-specific reweighting, threshold adjustment, or post hoc selection between equal and unequal fusion. Because test set 1 and test set 2 had been observed during earlier project development, this statement describes the formal current protocol rather than a claim that the benchmarks were historically blinded.

The experts agreed on most benchmark peptides, but the discordant samples revealed a reproducible pattern of complementary error correction: the structural expert provided correct counter-evidence for more than one third of the sequence expert's errors (Figure 3B). This rescue was strongly class-asymmetric, with the structural expert mainly correcting false-positive predictions of the sequence expert, whereas improvements in toxic-peptide recall were limited. Peptides classified correctly only by the 3D expert also tended to be shorter, but none of the prespecified attribute comparisons remained significant after Benjamini–Hochberg correction, so this length pattern is descriptive rather than evidence that the structural expert is generally superior for short peptides.

These results support a complementary rather than substitutive role for the structural expert (Figure 3).

3.3 Analysis of the Sequence and Structural Experts

The 1D expert is not the principal innovation of this work and is analyzed only briefly. Relative to the earlier pooled-ProtT5–TextCNN pipeline, the LoRA-adapted ProtT5 with bidirectional Mamba kept BACC and MCC broadly comparable on training OOF (0.9300 vs 0.9317 and 0.8562 vs 0.8586) while improving the ranking metrics (AUROC/AUPRC 0.9761/0.9533 vs 0.9724/0.9436), and OOF-fitted Platt calibration further reduced the Brier score on test set 1 and test set 2 (Table 4); the stronger ranking stream benefits the subsequent fusion with structural evidence (architecture in Section 2.3).

Descriptive structure-quality analysis found an association between mean pLDDT and subgroup performance in the historical benchmark data. A separate training-OOF comparison found that hard exclusion of low-confidence samples reduced full-domain performance, supporting the retained continuous-gating design without establishing pLDDT as a biological mechanism. In parallel, a Cα-level residue graph compresses chemically complex side chains into single nodes, whereas a paired five-fold OOF experiment showed that adding the full-atom chemical-site residual to the frozen backbone increased the BACC point estimate from 0.9153 to 0.9171 and the MCC point estimate from 0.8312 to 0.8342 (Table 7, bottom rows). These modest internal point-estimate changes are consistent with complementary chemical information but do not by themselves establish a general performance gain. Local geometric evidence is meaningful only in the appropriate sequence context: the same distance or hydrophobic contact can have different implications depending on residue identity and position, and the model therefore uses pLDDT-aware cross-attention to let sequence states selectively retrieve spatial evidence, with node- and peptide-level confidence gating controlling how much structural context is injected (mechanisms in Section 2.4).

A single predicted structure provides only one deterministic anchor, whereas short peptides may exhibit conformational heterogeneity [15]. GHXTox compresses conformer ensembles into statistics and allows them to produce only a zero-initialized logit residual. On training OOF, activating the multi-source residual increased MCC from 0.87255 to 0.87445 (paired-bootstrap $\Delta = +0.00190$, 95% CI $[-0.00043, 0.00454]$, $p = 0.203$). Under the frozen test protocol, activating the residual on the covered records changed the metrics negligibly (Tables 5 and 6: test set 1 MCC 0.8705 to 0.8726 and test set 2 MCC 0.6673 to 0.6746, with about one prediction flip on each benchmark), which is consistent with the decision not to enable conformer statistics during benchmark inference. This pattern is consistent with, but does not establish, a limitation in mapping conformational variation to toxicity labels; the present ablation cannot isolate that possibility from conformer quality, coverage, or residual-model capacity.

After combining all components, the 3D-v2 structural expert reached BACC, MCC, AUROC, and AUPRC values of 0.9208, 0.8464, 0.9711, and 0.9474 on test set 1, compared with 0.9179, 0.8458, 0.9699, and 0.9426 for the historical Cα-only 3D-v1, a consistent but modest improvement; on test set 2, AUPRC increased from 0.6969 to 0.7241 whereas MCC decreased from 0.6320 to 0.6128 (Table 7 and Figure 4). Finer-

grained structural modeling therefore improves positive-class probability ranking but does not guarantee simultaneous gains in all metrics at a fixed operating point.

These ablation and comparison analyses are based on the frozen historical test set 1 and test set 2 benchmarks and are therefore retrospective evidence. Test set 2 contains only 46 toxic peptides, so SN, BACC, and MCC are sensitive to a few threshold-adjacent samples, which partly explains the dataset-dependent direction of the 3D-v2 gains; the multi-conformer test ablation used the archived test-side conformer cache (486 and 284 covered records on test set 1 and test set 2, with fixed source and sequence-matching policies) and activated the residual only on the covered subset, so its negligible effect should not be extrapolated as evidence that conformer information is useless.

3.4 Robustness, Statistical Uncertainty, and Applicability Analysis

Point estimates alone are insufficient for distinguishing a stable model improvement from variation caused by finite samples. We therefore use multi-seed averaging for the 1D expert and paired, sample-level bootstrap resampling for model comparisons, all performed on the frozen predictor without retraining, recalibration, or threshold adjustment. Three full-data 1D models (seeds 42, 123, and 2025) are averaged before calibration; this ensemble is the final predictor, with no seed selected according to test set 1 or test set 2 performance, and a 15-member fold-checkpoint ensemble was examined but not adopted because it did not improve the frozen validation criteria.

For review-oriented screening, a reject-option policy [40] was defined using the uncertainty-ranking score given by the decision-boundary proximity $d_i = 1 - |p_i - \tau|/\max(\tau, 1-\tau)$ with $\tau = 0.47022256531284556$. On training OOF this proximity provides an informative ranking: as coverage decreases from 100% to 80%, accepted-sample error decreases and MCC increases monotonically, whereas equal-count random rejection shows no comparable improvement; at the prespecified 90% operating point the cutoff was frozen at $d^*_{\mathrm{OOF}} = 0.47361765874159856$ (Figure 5). Transferring this cutoff unchanged to test set 1 and test set 2 reduced the accepted-sample error by about 2.2–2.5 percentage points and improved MCC and BACC, while equal-count random rejection remained close to the full-coverage error (Table 8). The transfer uses no benchmark-specific quantile, coverage, class prevalence, or label, so the naturally different realized coverages reflect a single OOF-derived cutoff rather than retuned abstention rates. This supports a review-oriented screening workflow in which boundary-proximal predictions are flagged for review, although it does not constitute independent validation, and the proximity score is a ranking statistic rather than a calibrated probability of prediction error. The original binary prediction remains available for every abstained peptide.

3.5 Interpretability and Representative Case Studies

The interpretation experiments used a prespecified case-selection rule: within each test set and each expert-outcome group, one toxic and one non-toxic peptide nearest the fusion threshold were chosen (ties resolved by source index), forming a 16-peptide cohort. Before perturbation, baseline probabilities were recomputed and compared with the stored prediction vectors; the maximum absolute discrepancy for the 1D expert within the case cohort was 3.45×10^{-4}, and the corresponding 3D discrepancies were 5.49×10^{-4} on test set 1 and 1.14×10^{-5} on test set 2, none of which changed any

decision. The interpretation analyses therefore describe model sensitivity on the frozen predictor rather than confirmed biological mechanisms.

Mean pLDDT was strongly associated with the learned global structural gate (Spearman 0.960 on test set 1, 0.950 on test set 2, pooled 0.957), indicating that the model assigns less structural weight when predicted-structure confidence is lower. This is a model-use diagnostic and is not direct evidence of peptide flexibility, disorder, or conformational heterogeneity. Across 445 individual site deletions, most changes were small (median 0.00048, mean 0.00997, 95th percentile 0.057, maximum 0.2994 in p_{3D}). In the separate category-level analysis, all sites carrying a named category were deleted together in each peptide (121 peptide–category perturbations). The mean absolute changes in p_{3D} were 0.126 for hydrophobic-category deletion, 0.0687 for aromatic-category deletion, and 0.00179 for sulfur-category deletion. These category-level sensitivities do not describe individual-site effects and should not be interpreted as causal toxicophores. The opposing directions of the largest deletion effects (Phe8 increased p_{3D}, Pro19 decreased it) show that deletion scores depend on the surrounding peptide context and cannot be read as an intrinsic toxicity label for a functional group (Figure 6).

Fixed late fusion agreed on most benchmark peptides, and the discordant cases illustrate its operational consequences (the aggregate pattern is reported in Section 3.2). The toxic peptide `sequence_23_23` (ACLDIGNSCRED, two cysteines) had a weak 1D probability of 0.2806 but a high-confidence 3D probability of 0.9954; fusion raised the probability to 0.4808 and restored the correct toxic classification. At the model level, this example shows that a cysteine-associated structural pattern supplied counter-evidence to a weak 1D score; it does not establish a cysteine-mediated biological mechanism. The toxic peptide `sequence_34_34` had a correct 1D probability of 0.5447 but a 3D probability of 3×10^{-5}, and fusion reduced the probability to 0.3922, producing a false negative; strong sequence evidence can therefore be overridden when the structural expert supplies sufficiently strong opposing evidence, and such single-sample evidence conflicts warrant further inspection (Figure 7). These cases illustrate the operational premise and failure modes of the model: its 1D and 3D scores can agree or conflict, and fixed late fusion can either correct or introduce an error. They do not confirm a causal biological role for any particular structural feature.

3.6 Limitations and Future Directions

The most fundamental limitation stems from the single-output nature of three-dimensional conformation prediction. Short peptides in solution typically populate multiple interconverting conformations, and toxicity is usually mediated by one particular conformation or class of conformations; a single predicted structure cannot cover this multimodal distribution. This is precisely why the project introduced multiple conformation generators (PepFlow and AlphaFold2/ColabFold, assigned by length range). However, different generators exhibit inherent noise, and normalizing conformers across models and sources further amplifies this inconsistency: activating the multi-conformer residual on the test sets produced negligible and inconsistent effects that did not justify a public inference protocol, and source-independent and fusion-aware candidates also failed their prespecified retention gates (Section 3.3, Tables 5 and 6), so the conformer residual was eventually disabled for benchmark and public inference and the structural

information-extraction capacity of the model was not fully released. On training OOF the residual improved MCC by only +0.00190 ($p = 0.203$, not significant), and expanding the conformer cohort from 250 to 761 peptides did not pass the retention gates because the added records were all negative. In addition, full-atom conformational sampling (e.g., PepFlow) is computationally far more expensive than single-structure prediction, which limited large-scale generation and deployment. We therefore hope that next-generation protein structure prediction models can provide, at acceptable computational cost, multiple possible conformations and their probabilities (for example, lightweight multi-conformer sampling based on flow matching or diffusion), more closely matching the multimodal nature of peptide conformation and potentially further releasing the value of conformational information in toxicity prediction.

The second limitation is that the 1D sequence and 3D conformation dimensions are combined with a fixed 0.72/0.28 weight rather than dynamically adjusted per sample. This may partly stem from the first limitation: the inherent noise among different conformation predictors masks the information that would indicate which dimension should be trusted for a given sample, and when structural evidence is itself inconsistent across generators, the model cannot reliably distinguish, per sample, the reliability of sequence evidence from that of conformational evidence and therefore falls back to fixed weights. Consistent with this, sample-level confidence gates and dynamic routing attempts did not pass the retention gates: the former produced MCC values below the fixed default, and the latter improved ranking and calibration but did not exceed the frozen default's decision metrics (Section 3.4). This reflects that the two dimensions have not yet achieved sufficiently effective information complementarity; adaptive and interpretable fusion based on evidence reliability, together with more consistent conformational evidence sources, are directions for future work.

4 CONCLUSIONS

GHXTox explores the complementary joint use of sequence and conformational evidence by combining task-adaptive LoRA-ProtT5–Bi-Mamba sequence expert with a pLDDT-aware multi-scale structural expert through probability-level fusion; the two experts are independent and their complementarity is characterized by the structural expert correcting a subset of sequence errors in a class-asymmetric way (Section 3.2). The structural expert integrates residue geometry, full-atom-derived chemical sites, and sequence-conditioned structural evidence. A separate multi-conformer residual was trained with the single-structure backbone frozen and is retained only for controlled ablations. The final predictor disables that residual and uses the structural expert's single-structure base probability. The 1D/3D weights (0.72/0.28) and the decision threshold (0.47022256531284556) were frozen from training OOF predictions and applied in the current evaluation without benchmark-specific retuning; test set 1 and test set 2 are historical benchmarks, not newly blinded external sets.

On the frozen historical test set 1 and test set 2 benchmarks, the model reaches MCC values of 0.8705 and 0.6673 and AUPRC values of 0.9674 and 0.8046, and the final fusion improves AUROC, AUPRC, and the Brier score relative to the calibrated 1D stream. The 3D expert correctly classified 36.92% and 37.50% of the respective test set 1 and test set 2 errors made by the 1D expert, with most of these structural rescues

occurring among non-toxic peptides. An OOF-frozen selective policy further reduced accepted-sample error to 2.71% and 3.16% at realized coverages of 88.54% and 86.94%, respectively, without benchmark-specific cutoff selection. Conformer-derived information produced only modest and non-uniform improvements, indicating that mapping conformational variation to toxicity-relevant evidence remains an important limitation. The interpretability analyses describe model sensitivity rather than causal mechanisms.

Several limitations remain, most notably the single-output nature of conformation prediction and the fixed probability-level fusion between the two dimensions, as discussed in Section 3.6; both benchmarks were also observed during earlier model development, and the structures are predicted rather than experimental. Looking ahead, we hope that more powerful three-dimensional conformation models, capable of producing multiple conformations with their probabilities at acceptable computational cost, together with dynamic, evidence-reliability-aware fusion between the sequence and structural dimensions, will enable a deeper complementary joint use of the two types of information, with experimental validation where feasible.

Data and Software Availability

The source code, configuration files, fold manifests, and inference scripts used in this study are available at `https://github.com/Suiyumanbu/GHXTox`. A versioned archival release containing the exact commit identifier, environment manifest, frozen model parameters, calibration coefficients, fusion weights, full-precision decision threshold, and sample-level prediction files will accompany the final publication. Raw peptide records were obtained from the public databases described in Section 2.1. Where redistribution is restricted by the original database terms, accession lists and preprocessing scripts are provided instead of redistributing the source records.

Author Contributions

All authors contributed equally.

Acknowledgements

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Conflict of Interest Disclosure

The authors declare no conflicts of interest.

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Table Captions

Table 1. Statistics of the datasets used in this study.
Table 2. Performance comparison on benchmark test set 1 and test set 2.
Table 3. Ranking and probability quality of the 1D streams and probability fusion. Structural probabilities use the canonical single-structure output.
Table 4. Controlled comparisons for the task-adaptive 1D expert. Training-OOF architecture and benchmark-calibration rows use separate protocols.
Table 5. test set 1 conformer-residual ablation on the full benchmark; the residual is activated only for the 486 covered peptides.
Table 6. test set 2 conformer-residual ablation on the full benchmark; the residual is activated only for the 284 covered peptides.
Table 7. Structural-expert ablations. Benchmark comparisons and paired training-OOF chemical-site rows use separate protocols.
Table 8. OOF-frozen selective prediction on test set 1 and test set 2. Random-rejection values average 5,000 equal-count masks.

Table 1. Statistics of the datasets used in this study.

Dataset	Toxic	Non-toxic	Total	Use
Training (benchmark)	1,818	4,569	6,387	Model training
test set 1 (benchmark)	320	806	1,126	Benchmark test
test set 2	46	536	582	Benchmark test

Table 2. Performance comparison on benchmark test set 1 and test set 2.

Dataset	Model	ACC	BACC	SN	SP	MCC	AUROC	AUPRC
test set 1	ToxMSRC	0.9405	0.9217	0.8781	0.9653	0.8520	0.9650	NR
	ToxPLTC	0.9378	0.9302	**0.9125**	0.9479	0.8496	0.9709	0.9426
	GHXTox (current)	**0.9476**	**0.9323**	0.8969	**0.9677**	**0.8705**	**0.9811**	**0.9674**
test set 2	ToxMSRC	0.9416	0.8689	0.7826	0.9552	0.6550	0.9430	NR
	ToxPLTC	0.9261	**0.8804**	**0.8261**	0.9347	0.6197	0.9295	0.7353
	GHXTox (current)	**0.9467**	0.8618	0.7609	**0.9627**	**0.6673**	**0.9433**	**0.8046**

Table 3. Ranking and probability quality of the 1D streams and probability fusion. Structural probabilities use the canonical single-structure output.

Dataset	Probability stream	AUROC	AUPRC	Brier
test set 1	1D, uncalibrated	0.9799	0.9644	0.0445
	1D, OOF Platt calibrated	0.9799	0.9644	0.0429
	Current fusion (0.72/0.28)	**0.9811**	**0.9674**	**0.0410**
test set 2	1D, uncalibrated	0.9409	0.7831	0.0491
	1D, OOF Platt calibrated	0.9409	0.7831	0.0451
	Current fusion (0.72/0.28)	**0.9433**	**0.8046**	**0.0436**

Table 4. Controlled comparisons for the task-adaptive 1D expert. Training-OOF architecture and benchmark-calibration rows use separate protocols.

Evaluation	Variant	BACC	MCC	AUROC	AUPRC	Brier
Training OOF	Pooled ProtT5–TextCNN	0.9317	0.8586	0.9724	0.9436	NR
Training OOF	LoRA-ProtT5–Bi-Mamba	0.9300	0.8562	**0.9761**	**0.9533**	NR
test set 1	Three-seed 1D, uncalibrated	–	–	0.9799	0.9644	0.0445
test set 1	Three-seed 1D, OOF Platt	–	–	0.9799	0.9644	**0.0429**
test set 2	Three-seed 1D, uncalibrated	–	–	0.9409	0.7831	0.0491
test set 2	Three-seed 1D, OOF Platt	–	–	0.9409	0.7831	**0.0451**

Table 5. test set 1 conformer-residual ablation on the full benchmark; the residual is activated only for the 486 covered peptides.

Variant	ACC	BACC	MCC	AUROC	AUPRC	Brier
Base (zero conformer)	0.9476	0.9323	0.8705	0.9811	0.9674	0.0410
+ Conformer residual	0.9485	0.9329	0.8726	0.9811	0.9675	0.0407

Table 6. test set 2 conformer-residual ablation on the full benchmark; the residual is activated only for the 284 covered peptides.

Variant	ACC	BACC	MCC	AUROC	AUPRC	Brier
Base (zero conformer)	0.9467	0.8618	0.6673	0.9433	0.8046	0.0436
+ Conformer residual	0.9485	0.8627	0.6746	0.9433	0.8043	0.0430

Table 7. Structural-expert ablations. Benchmark comparisons and paired training-OOF chemical-site rows use separate protocols.

Evaluation	Variant	BACC	MCC	AUROC	AUPRC
test set 1	Historical Cα 3D-v1	0.9179	0.8458	0.9699	0.9426
test set 1	pLDDT-aware multi-scale 3D-v2	**0.9208**	**0.8464**	**0.9711**	**0.9474**
test set 2	Historical Cα 3D-v1	**0.8481**	**0.6320**	**0.9316**	0.6969
test set 2	pLDDT-aware multi-scale 3D-v2	0.8453	0.6128	0.9295	**0.7241**
Training OOF	Frozen backbone, no chemical-site residual	0.9153	0.8312	0.9687	0.9388
Training OOF	Frozen backbone + chemical-site residual	**0.9171**	**0.8342**	**0.9692**	**0.9401**

Table 8. OOF-frozen selective prediction on test set 1 and test set 2. Random-rejection values average 5,000 equal-count masks.

Dataset	Method	Coverage (%)	Error (%)	MCC	BACC	SN	SP
test set 1	Full binary prediction	100.00	5.24	0.8705	0.9323	0.8969	0.9677
	Frozen selective prediction	88.54	2.71	0.9329	0.9629	0.9397	0.9860
	Equal-count random rejection	88.54	5.24	0.8705	0.9323	0.8969	0.9677
test set 2	Full binary prediction	100.00	5.33	0.6673	0.8618	0.7609	0.9627
	Frozen selective prediction	86.94	3.16	0.7781	0.8983	0.8158	0.9808
	Equal-count random rejection	86.94	5.33	0.6670	0.8617	0.7607	0.9627

Figure Captions

Figure 1. GHXTox framework. The peptide sequence follows parallel sequence and structure pathways. The sequence pathway uses ProtT5 tokenization for the 1D model, whereas the structure pathway uses multi-conformer training and single-structure inference with pLDDT-aware structural inputs for the 3D model.

Figure 2. GHXTox discrimination and probability reliability. (A–C) test set 1 ROC, precision–recall, and reliability curves; (D–F) corresponding test set 2 curves. Orange, blue, and navy denote the task-adaptive 1D expert, the 3D structural expert, and probability fusion, respectively. Gray dashed lines denote chance, prevalence, or ideal-reference behavior.

Figure 3. Complementarity of the 1D and 3D experts on test set 1 and test set 2. (A) Expert-outcome groups and their toxic fractions. (B) Conditional rescue rates with 95% Wilson intervals. (C) Prespecified 3D-only versus 1D-only attribute effects with 95% bootstrap intervals; none remained significant after Benjamini–Hochberg correction.

Figure 4. Protocol-separated structural ablations. (A, B) Previous and pLDDT-aware 3D experts on test set 1 and test set 2. (C) Training-OOF comparison without and with chemical-site information.

Figure 5. OOF-frozen selective prediction and benchmark transfer. (A, B) Training-OOF accepted error and MCC across coverage levels. (C–E) Realized coverage, accepted error, and MCC after applying the unchanged cutoff to test set 1 and test set 2. Shading and error bars denote the manuscript-defined 95% bootstrap intervals; no benchmark-specific cutoff was recomputed.

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Figure 7. Representative expert disagreements. (A) Structural evidence rescues a toxic decision missed by the 1D expert. (B) Opposing structural evidence reverses a correct 1D decision. Bars show expert and fused probabilities relative to the frozen threshold; the cases are illustrative, not causal biological evidence.

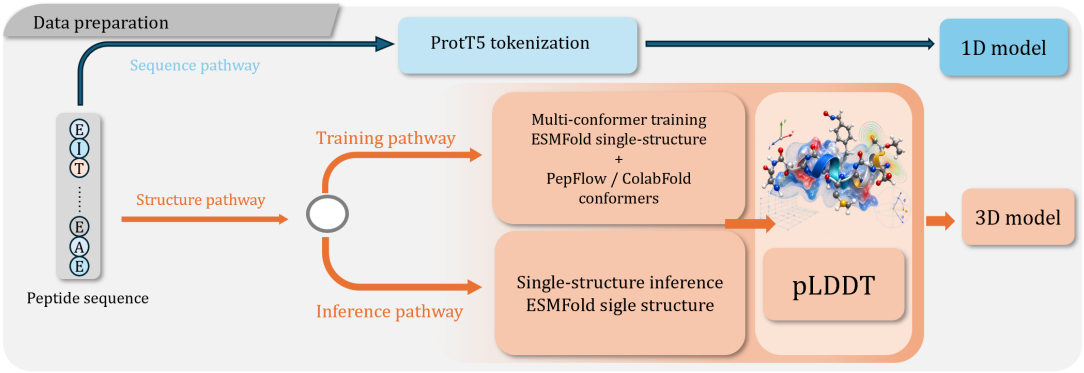


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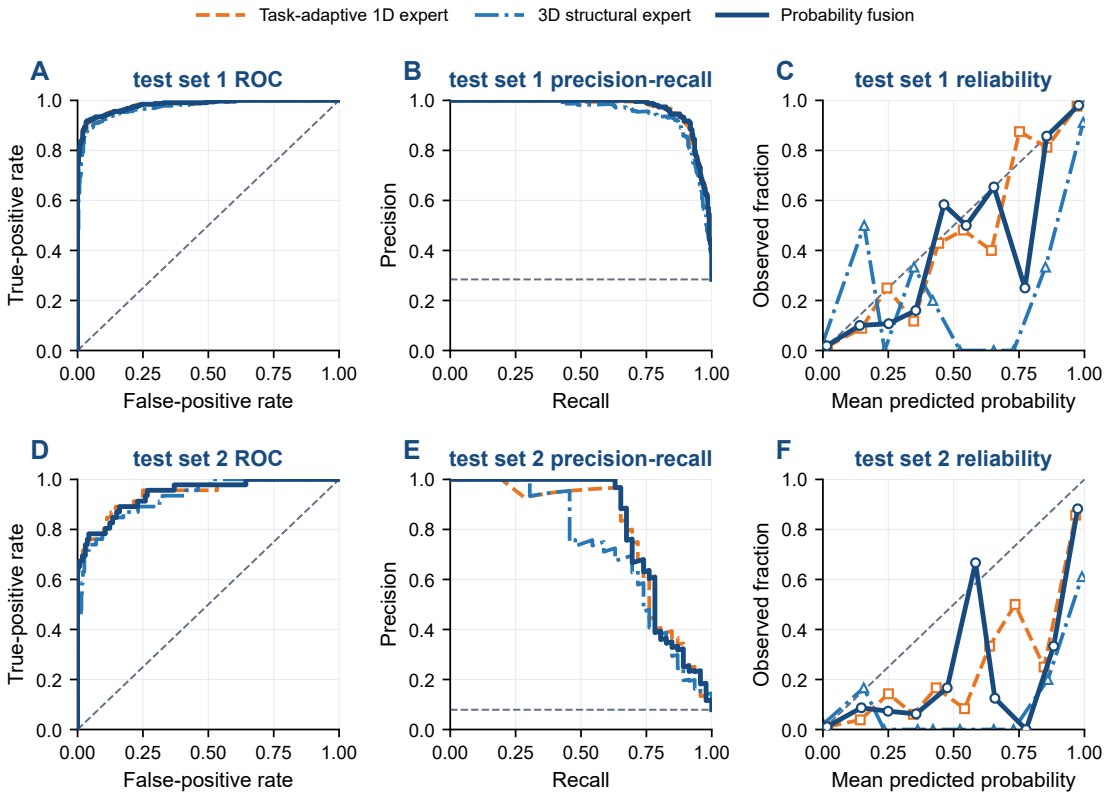


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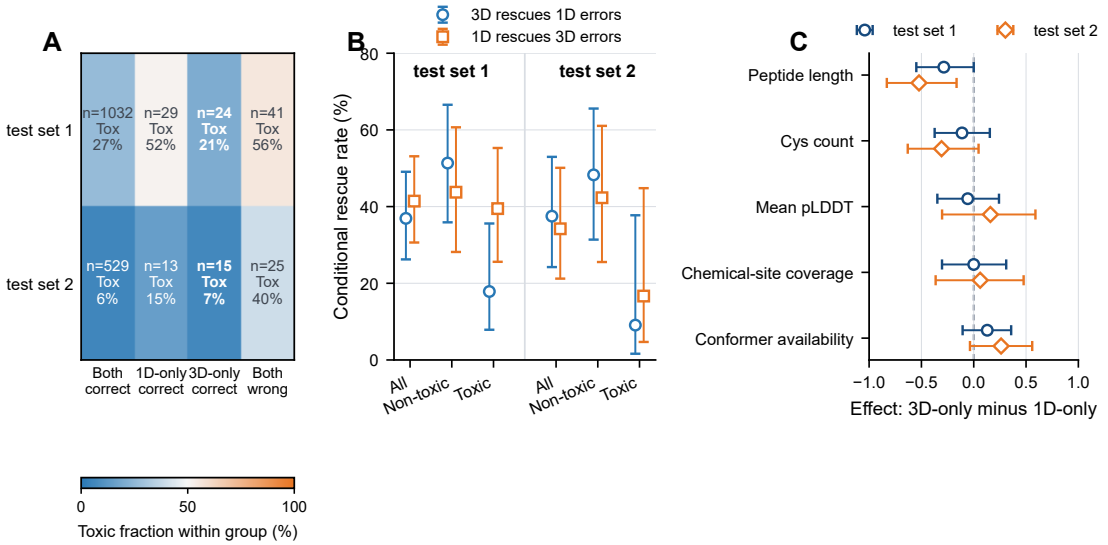


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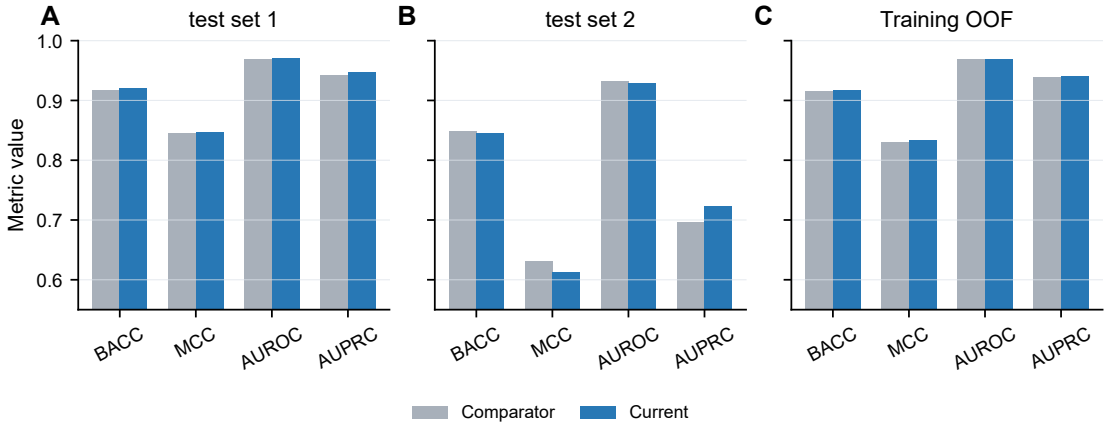


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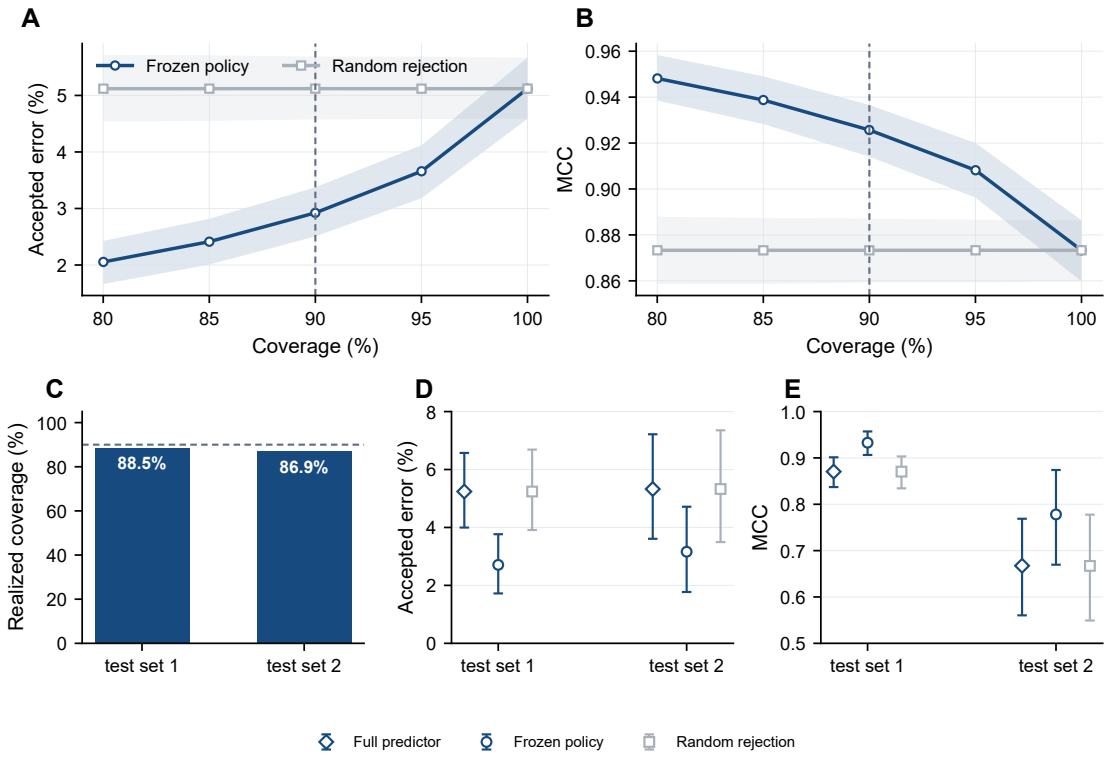


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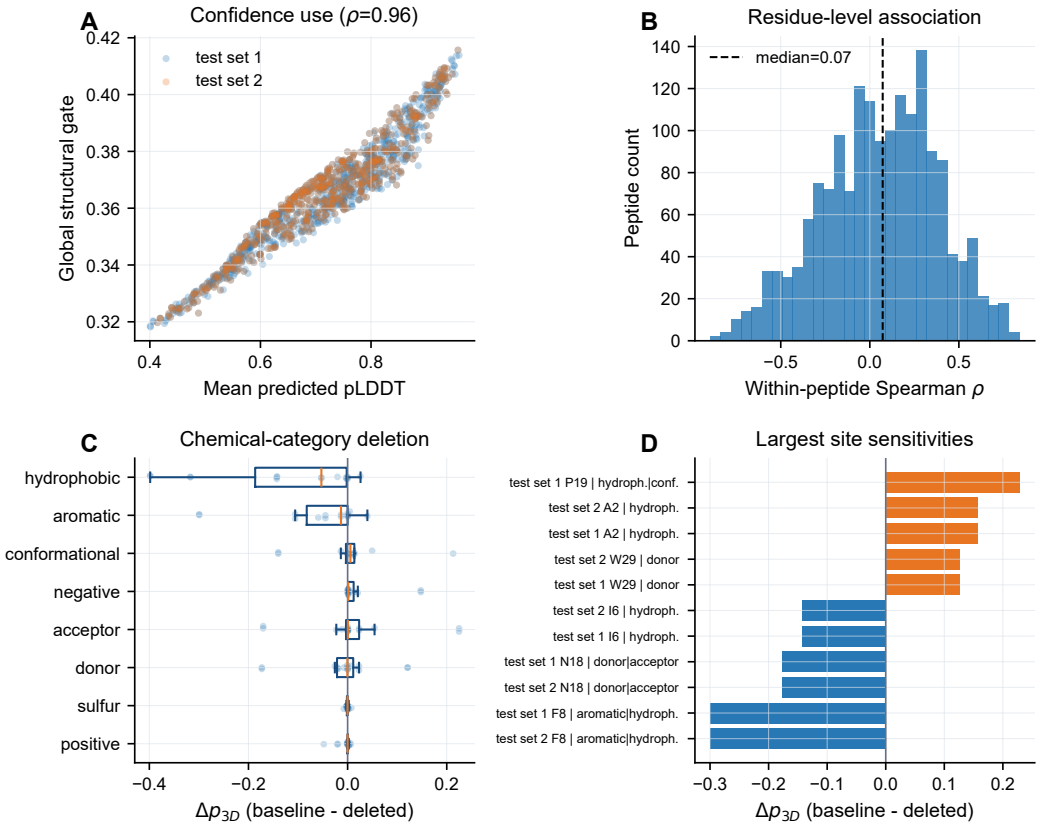


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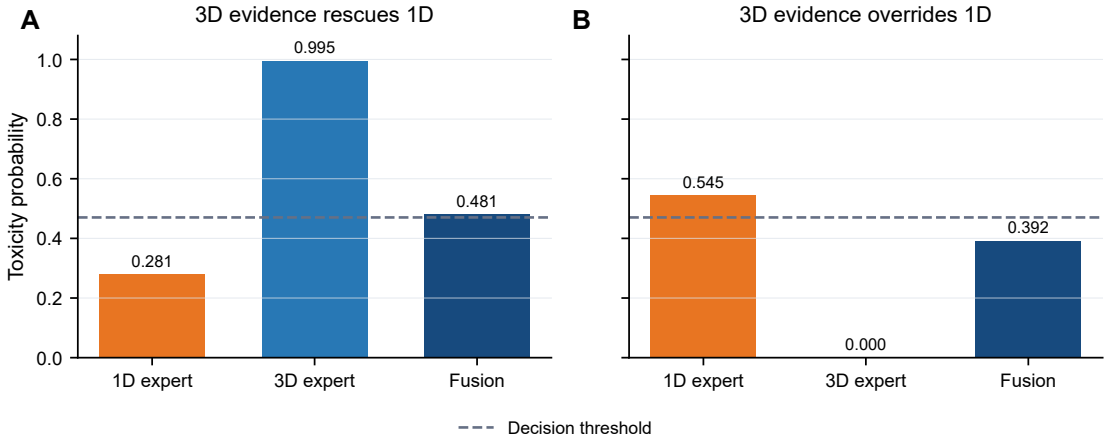


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